

SPECTRAL PROBING OF FROZEN AUDIO ENCODERS: AN EXPLORATORY STUDY OF LOW-FREQUENCY CONCENTRATION

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ABSTRACT

Music has inherent hierarchical structure and periodicity across multiple timescales, making spectral decomposition a natural lens for studying audio representations. This report documents an exploratory study of where task-relevant information lives along the time axis of frozen audio encoders, viewed through the lens of Fourier decomposition along the temporal dimension. Inspired by spectral probing in NLP, we initially expected the dominant frequency band of a per-band linear probe to track the timescale of the downstream task: slowly varying labels (genre, key) should be readable from low temporal-frequency bands, and rapidly varying labels (per-frame chord) from higher bands. This hypothesis was wrong. Across four frozen encoders (CLAP, AudioMAE, BEATs, PaSST) and four MIR benchmarks (FMA-Small genre, NSynth instrument family, GuitarSet key, GuitarSet per-frame chord), the best per-band probe lands at or near the lowest band and the band ranking is essentially independent of task timescale; per-band accuracy decays non-monotonically with a few small interior bumps but no competitively performing high band. We name the phenomenon *spectral concentration in representation*. Four null perturbations on PaSST (random-init encoder, label shuffle, DC removal, time permutation) confirm the effect is not a probe artifact, not a DC artifact, partly architectural, and partly tied to temporal ordering. The *cause* of the concentration remains open: we end with three non-exclusive working hypotheses—an architectural inductive bias, a property of classification labels as temporal averages, and a training-dynamics account suggested by the well-known spectral bias of neural networks, but not yet testable here because no non-low band probes competitively enough to anchor a training intervention—together with the experiments needed to disentangle them.

1. INTRODUCTION

This is a final report on an exploratory study of where task-relevant information lives along the time axis of frozen audio encoders, inspired by spectral probing of contextualised embeddings in NLP [1]. By “frequency” we mean the rate of change of the encoder’s representation function as the input varies along the time axis, in the sense of [1]:

each embedding dimension is treated as a sequence in time, and a low (resp. high) frequency band corresponds to a slowly (resp. rapidly) varying component—not an acoustic spectrogram or a learned spectral head over word vectors. Linear probing of frozen audio encoders is markedly less reliable than its computer-vision counterpart and tends to under-report what the encoder actually represents on multi-label tasks [2]; the same probing tool has nevertheless been used to recover non-trivial cross-modal signal from frozen audio embeddings, e.g. to test whether text-only language models encode grounded sound knowledge [3]. We use it here as a coarse instrument and redirect it onto the *temporal spectrum* of the embedding representation.

We approached the question with two pre-registered hypotheses. **H1 (frequency–task alignment)**. The dominant band of a per-band probe should track the timescale of the downstream task: clip-level labels (genre, key) reading out of low bands, fast-varying labels (per-frame chord) reading out of higher bands—the audio analogue of the band-by-task specialisation reported for NLP [1]. **H2 (spectral-bias-driven redistribution)**. Conditional on H1 failing in a specific way—some non-low band probing competitively against the lowest—we planned a training-time intervention that exploits the well-known spectral bias of neural networks [4,5] to redistribute task-relevant information across bands. Such interventions are routine in vision and graphics: NeRF added a sinusoidal positional encoding to input coordinates expressly to overcome MLP spectral bias [6]; a Fourier-features analysis [7] subsequently re-cast this as random Fourier features and explained the effect via the neural tangent kernel; and explicit high/low decoupling continues to improve video neural representations [8]. The audio analogue has a clear optimisation target only when some non-low band already shows non-trivial task signal the linear probe reads as under-exploited; absent that, no obvious training-time redistribution exists.

We probed four frozen encoders (CLAP [9], AudioMAE [10], BEATs [11], PaSST [12]) on four MIR tasks spanning timescales from track-level (FMA-Small [13] genre, GuitarSet [14] key) to note-level (NSynth [15] instrument family) to per-frame (GuitarSet chord detection). H1 fails outright: per-band probe accuracy departs sharply from any high-frequency-for-fast-tasks ranking, and on the per-frame chord task, where H1 predicted high-band dominance, the gap between the lowest band and the rest is in fact the largest of all four tasks (Figure 1d). The decay is also non-monotone: most curves drop sharply from band 0

Work in progress.

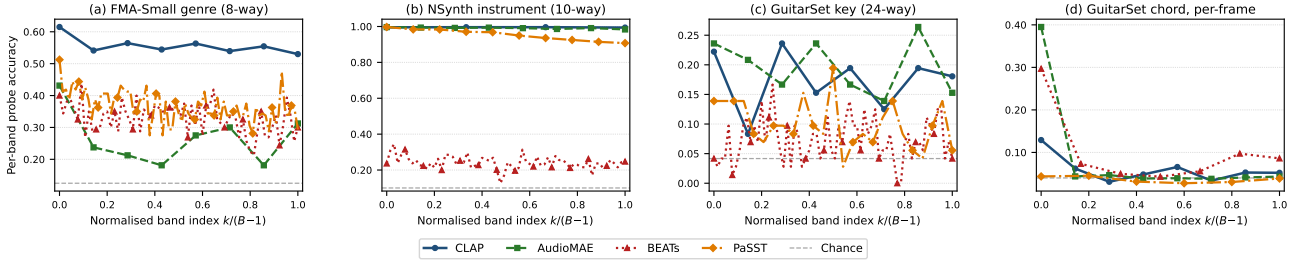


Figure 1. Per-band linear-probe accuracy vs. normalised temporal-band index $k/(B-1)$. Panels: (a) FMA-Small: genre classification; (b) NSynth: instrumental identification; (c) GuitarSet: global key identification (major/minor); (d) GuitarSet: per-frame chord classification. Encoders: CLAP, AudioMAE, BEATS, PaSST. Horizontal dashed: chance level.

to band 1 and then oscillate, with secondary local peaks that on a few (encoder, task) pairs slightly exceed band 0. However, none of these interior bumps exposes a non-low band the probe reads as under-exploited, so H2’s intervention had no optimisation target and we did not pursue it; this is not a refutation of spectral-bias-as-cause, since the high-band information may be non-linearly encoded or genuinely absent and our linear probes cannot tell those apart.

We name the phenomenon *spectral concentration in representation*, verify that it survives four null perturbations on PaSST (random-init encoder, label shuffle, DC removal, time permutation; Section 4), and end the report with three non-exclusive working hypotheses for its origin—architectural inductive bias, training-dynamics-driven concentration we cannot yet test, and a task-timescale account in which long-horizon labels are simply low-frequency in the representation—together with the experiments we believe should be tried next (Section 5).

2. SPECTRAL BAND PROBING

Spectral probing in NLP used the DCT [1]; the DCT’s even-symmetric extension is a poor fit for encoder activations that are neither periodic nor symmetric. We use the magnitude of the one-sided real FFT (orthonormal `rfft`), which provides a uniform frequency grid compatible with fixed-width band partitioning. Using the full complex spectrum is a straightforward extension we leave for future comparison.

2.1 FFT along the time axis

Let $x \in \mathbb{R}^{F \times T}$ denote a frozen encoder’s output for one input, where F is the number of feature channels (latent dimensions) and T is the number of time frames. We apply an orthonormal one-sided real DFT along the time axis, independently for each channel $f \in \{1, \dots, F\}$:

$$X_f[k] = \frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} x_f[t] e^{-i 2\pi kt/T}, \quad (1)$$

$$k = 0, \dots, K-1,$$

with $K = \lfloor T/2 \rfloor + 1$ (e.g., $T=64 \Rightarrow K=33$). As features for probing we use the magnitude spectrogram of the

representation,

$$M_{f,k} = |X_f[k]|, \quad M \in \mathbb{R}^{F \times K}, \quad (2)$$

so that each sample is summarised by a $(F \times K)$ matrix of how strongly each latent dimension modulates across the T frames. No window is applied by default, matching the rectangular-window setting of our main experiments.

2.2 Fixed-band filters

We partition the K DFT bins into B non-overlapping, equal-width bands. Given a band width W ,

$$B = \lfloor K/W \rfloor, \quad \mathcal{I}_b = \{bW, bW+1, \dots, (b+1)W-1\}, \quad (3)$$

for $b = 0, \dots, B-1$; bins beyond BW are discarded. Let $M[:, \mathcal{I}_b] \in \mathbb{R}^{F \times W}$ denote the sub-matrix of M whose columns are indexed by $\mathcal{I}_b$, i.e. the rows stay as the full set of F channels while the columns restrict to the W frequency bins of band b . We form the band- b feature by flattening this sub-matrix into a single vector,

$$\phi_b(x) = \text{vec}(M[:, \mathcal{I}_b]) \in \mathbb{R}^{F \cdot W}, \quad (4)$$

where $\text{vec}(\cdot)$ stacks the columns of $M[:, \mathcal{I}_b]$ in order, each column in $\mathbb{R}^F$, to form a vector in $\mathbb{R}^{FW}$. For each band b independently, we fit a multinomial softmax logistic-regression probe

$$p_b(y|x) = \text{softmax}(W_b \phi_b(x) + c_b), \quad (5)$$

with $W_b \in \mathbb{R}^{C \times (FW)}$ and $c_b \in \mathbb{R}^C$, over the C task classes. All downstream benchmarks in this paper are multi-class (genre, instrument family, and key), so each probe uses the same softmax head with task-dependent C . Training uses ℓ_2 -regularised logistic regression on one encoder’s frozen features; no fine-tuning of the encoder occurs.

Why uniform, not Mel, bands? The bin index k in Eqn (1) counts how rapidly each latent dimension oscillates over frame index, and f indexes learned channels rather than acoustic frequency, so Mel-style perceptual warping has no direct analogue here; we keep uniform width- W bands on the native DFT grid.

Model	FMA_Small (Genre)					NSynth (Instrument)					GuitarSet (Key Det.)					GuitarSet (Chord Det.)				
	ORIG.	B _*	Top- <i>K</i>	Cumul.	SP	ORIG.	B _*	Top- <i>K</i>	Cumul.	SP	ORIG.	B _*	Top- <i>K</i>	Cumul.	SP	ORIG.	B _*	Top- <i>K</i>	Cumul.	SP
CLAP	0.616	0.615 _{0/8}	0.616	0.636*	0.639	0.997	0.996 _{1/4}	0.997	0.998	0.998	0.250	0.236 _{2/8}	0.181	0.278	0.250	0.105	0.129 _{0/8}	0.135***	0.138***	0.138***
AudioMAE	0.287	0.431 _{0/8}	0.362	0.431**	0.475***	0.994	0.997 _{0/8}	0.997	0.998*	0.997	0.375	0.264 _{6/8}	0.375	0.431	0.417	0.341	0.395 _{0/8}	0.353	0.395***	0.418***
BEATs	0.613	0.438 _{6/66}	0.550	0.662	0.606	0.854	0.345 _{2/66}	0.458	0.858	0.976***	0.194	0.167 _{16/66}	0.153	0.250	0.375*	0.282	0.297_{0/7}	0.296***	0.297***	0.261
PaSST	0.469	0.512 _{0/75}	0.556	0.556*	0.588*	0.989	0.997 _{0/10}	0.993	0.997***	0.998***	0.472	0.194 _{12/25}	0.319	0.514	0.292	0.036	0.045 _{1/6}	0.047***	0.049***	0.037

Table 1. Linear spectral-probe accuracy for four encoders (rows) across four benchmarks (column groups), each with five probe configurations. ORIG.: full-spectrum probe; B_{*}: best single band (subscript b/B : winning band out of B); Top-*K*: top-3 bands combined; Cumul.: best low-to-high prefix; SP: learnable spectral filter [1]. **Bold**: row maximum. Superscripts (* $p<0.05$, ** $p<0.01$, *** $p<0.001$): McNemar vs. ORIG., shown only when strictly greater; underline: maximum and $p<0.05$.

Null perturbations. To probe the origin of any band-wise structure we observe, we use four null perturbations of one encoder–task pair: (A) random-init encoder, all reset-capable submodules re-initialised before extraction, isolating the architectural slice from the learned slice; (B) label shuffle, training labels permuted, isolating probe leakage from genuine signal; (C) DC removal, zeroing coefficient 0 of $|rfft|$, isolating per-channel time-mean information from genuine low-frequency modulation; (D) per-sample temporal permutation, applied before the rFFT, isolating per-frame information content from temporal ordering. All four perturbations are applied to PaSST on FMA-Small and NSynth.

3. EXPERIMENTAL SETUP

We test four audio embeddings, one from each of four pretrained models—CLAP, AudioMAE, BEATs, and PaSST—and use them to probe four classification tasks spanning a hierarchy of audio time scales (track-level genre, track-level key, note-level instrument family, per-frame chord), drawn from three standard datasets. All encoders are frozen: the only trainable parameters in our experiments are those of the linear probes in Eqn (5). Full checkpoint identifiers, feature shapes, label inventories, and pool sizes are collected in Appendix A.

3.1 Models

CLAP is a contrastive audio–text encoder whose audio branch is the HTS-AT hierarchical audio transformer [9, 16]. AudioMAE is a ViT-B/16 spectrogram encoder pretrained with masked autoencoding on AudioSet-2M and fine-tuned on AS-20k [10]. BEATs is a self-supervised audio encoder trained with an acoustic tokenizer on AudioSet [11]. PaSST is a supervised AudioSet-tagging transformer trained with patchout regularisation [12]. Together the four cover the main paradigms for general-purpose audio embeddings—contrastive, masked-prediction, tokenizer-based self-supervision, and fully supervised tagging—by design, so that any spectral concentration that recurs across all of them is unlikely to stem from any one training objective alone.

3.2 Tasks

The four tasks are all closed-set multi-class classification and are chosen to span a hierarchy of audio time

scales, from track-level to per-frame. At the long-horizon end, FMA-Small genre is an eight-way classification over whole tracks, and GuitarSet key detection is a nineteen-way classification over the 19 root-mode labels that appear in the 360-track mono pickup-mix pool (drawn from the 24-way chromatic root $\times$ {major, minor} label space). At an intermediate horizon, NSynth instrument recognition classifies each isolated 4-second note into one of eleven instrument families. At the short-horizon end, GuitarSet chord detection is a per-frame 25-way classification (12 maj + 12 min + no-chord under the MIREX reduction of the JAMS chord annotations [17]) on the same 360-track pool. The same softmax probe of Eqn (5) is used for every task.

3.3 Datasets

The three datasets are all widely used benchmarks and were chosen so that each horizon of the task hierarchy is represented by a clean, well-labelled source. FMA-Small [13] supplies an eight-genre balanced subset of the Free Music Archive for long-horizon music labels that should be recoverable from slowly varying, track-level cues. NSynth [15] supplies a large collection of isolated instrument notes, stripping away long-range temporal structure and leaving only timbre. GuitarSet [14], through its synchronised JAMS key annotations [17], supplies a track-level tonal label over a uniform solo-guitar acoustic setting, controlling for timbre, microphone, and player identity.

4. RESULTS

We discuss the central finding (Section 4.1), the four null perturbations (Section 4.2), and a brief comparison with the spectral-probing auto selector (Section 4.3).

4.1 Spectral Concentration in Representation

The headline finding is in the B_{*} and Cumul. columns of Table 1 and visualised across the full per-band sweep in Figure 1. On most encoder $\times$ benchmark cells the best single width-4 band recovers most or all of the ORIG. accuracy. For several cells the winning band b_* sits at the very lowest index (PaSST FMA $b=0/75$, PaSST NSynth $b=0/10$, CLAP FMA $b=0/8$, AudioMAE FMA $b=0/8$, three of four chord cells); for the rest, b_* moves to a small interior index that still lies in the lower portion of the spectrum (CLAP GuitarSet key $b=2/8$, AudioMAE

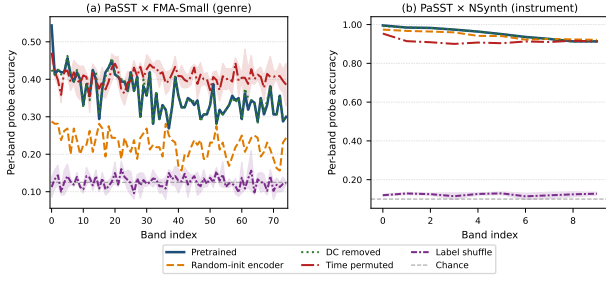


Figure 2. Per-band probe accuracy for PaSST under four null conditions alongside the pretrained baseline (3 seeds each; shaded: $\pm$ std). Left: FMA-Small. Right: NSynth. Dashed: chance.

GuitarSet key $b=6/8$, BEATs NSynth $b=2/66$, BEATs FMA $b=6/66$, PaSST GuitarSet key $b=12/25$), typically beating $b=0$ by a small margin. The decay is also non-monotone: most curves drop sharply from band 0 to band 1 and then oscillate with secondary local peaks (Figure 1), rather than falling off cleanly. AudioMAE on FMA-Small is the single most striking case: a single band (out of 8) lifts accuracy from 0.287 to 0.431, and the cumulative low-to-high prefix tops out at 0.475 (+18.8 pp over ORIG., $p < 10^{-3}$). The qualitative pattern—band-0 or a near-band-0 interior bump dominating, a sharp drop from band 0, then a non-monotone tail—holds across self-supervised, masked-prediction, contrastive, and supervised encoders, so it is unlikely to be tied to a specific training objective. Crucially for H1, the band ranking is also essentially independent of task timescale: on the per-frame chord task (Table 1, last columns; Figure 1d), where H1 predicted high-band dominance, three of four encoders place b_* at $b=0$ and the gap between the lowest band and the rest is the largest of all four tasks.

4.2 Null Perturbations

Four null perturbations on $\text{PaSST} \times \{\text{FMA-Small}, \text{NSynth}\}$ (Section 2) test alternative explanations. (B) Label shuffle collapses ORIG. to chance (0.163 ± 0.010 on FMA, 0.125 ± 0.006 on NSynth), ruling out probe leakage. (C) DC removal changes ORIG. accuracy by at most 0.007 and the lowest band still wins, ruling out the trivial “ B_0 = per-channel time-mean” objection. (A) Random-init encoder separates the two tasks: on near-stationary NSynth the random encoder reaches 0.978 (vs. 0.989 pretrained, $\Delta = -0.012$), so almost all of the low-band advantage is architectural; on temporally structured FMA-Small genre it drops to 0.288 (vs. 0.469, $\Delta = -0.181$), so $\sim 60\%$ of the advantage is learned. (D) Per-sample temporal permutation preserves total decodable information on FMA (0.500 ± 0.014) but the winning band becomes random across seeds (B_{61}, B_0, B_0); on NSynth it leaves the band structure intact (B_0 still wins, 0.954). Together (Figure 2), the nulls localise the effect: it is not DC, not a probe artifact, partly architectural (more so on stationary tasks), and partly a property of the encoder’s temporal ordering (more so on temporally structured tasks).

4.3 Auto Selectors Match or Exceed ORIG.

A subtler observation: **Cumul.** is $\geq$ ORIG. in 16/16 cells (sign test $p \approx 3.1 \times 10^{-5}$, mean +3.8 pp), and the spectral-probing auto **SP** [1] is $\geq$ ORIG. in 13/16 cells ($p \approx 0.021$). **Top- K** with fixed $K=3$ is inflexible and on average costs 3.0 pp. McNemar tests on the shared split [18] confirm that several **Cumul./SP** wins are individually significant, with the largest absolute gains on the low-resource cells (FMA-Small, GuitarSet) where the full-spectrum input dimension is several hundred times the number of training samples (e.g., $F \cdot n_{\text{coeffs}}/N_{\text{train}} \approx 609$ for FMA-Small/PaSST), so removing irrelevant high-frequency coefficients yields a better-conditioned probe. The Minimum Description Length (MDL) reading of probing [19, 20] fits this picture: a probe that uses fewer coefficients yet matches or exceeds full-spectrum accuracy is providing a more compact, not weaker, account of the task.

5. DISCUSSION

What follows are three non-exclusive working hypotheses for the origin of *spectral concentration in representation*, each paired with a concrete next-step experiment. We do not claim to choose between them in this report; the experiments below are what we believe should be tried next.

5.1 Hypothesis A: Architectural Inductive Bias

A frozen but randomly initialised PaSST already places the best single band at $b=0$ on both FMA-Small and NSynth, and on stationary NSynth it reaches 98% of the pretrained accuracy. This says architecture alone—tokenisation, attention, positional encoding—suffices to expose a usable low-frequency band, before any task-relevant training. Two candidate mechanisms are worth examining: (i) the implicit low-pass induced by patchout/striding in audio transformers [12], and (ii) the well-documented spectral bias of neural function classes towards low-frequency components [4], which here would manifest at the representation rather than the function level. Disentangling these would require swapping the front-end (e.g., MAE patchifier vs. HTS-AT swin) while holding training fixed, or reading the eigenstructure of the random-init covariance along the time axis.

5.2 Hypothesis B: Training Further Concentrates an Already-Biased Spectrum

On FMA-Small genre, $\sim 60\%$ of the random-init $\rightarrow$ pretrained accuracy gap closes only after training, and the band-vs-non-band margin widens. Training therefore appears to concentrate an architectural prior further, but the evidence is weak in two ways. First, our four encoders are checkpoints not full training histories: we cannot say whether concentration builds up steadily during training, settles in early, or rises and falls together with downstream accuracy. Second, the absence of a competitive non-low band on any task means we cannot use the standard linear-probe handle to test whether spectral

bias [4] is at play here—high-frequency content may be present in the representation but non-linearly encoded, in which case the linear probe is blind to it whether spectral bias holds or not.

The natural experiment is to retrain a small audio transformer from scratch on AudioSet (or a controlled subset), freeze checkpoints every ~ 5 epochs, and run the per-band probe on each checkpoint. Two summary numbers track how concentration evolves over training: the best-single-band accuracy and the fraction of ORIG. accuracy recovered by the lowest band. A control run with shuffled labels, under the same schedule, separates label-driven concentration from concentration that the training procedure produces on its own. This would directly test whether spectral concentration in representation is a consequence of spectral bias during training.

5.3 Hypothesis C: The Task Itself Lives at a Long Timescale

Genre, instrument family, and key are all clip-level labels that change on timescales of seconds to tens of seconds, and these are the tasks on which the lowest band wins outright ($b=0$ in the B_* column for almost every cell). This matches mainstream MIR practice, where mean pooling (the $W=1$ DC band) has remained competitive against more elaborate filtering and feature-aggregation schemes. A simple reading of our results, then, is that long-horizon labels are low-frequency in the representation, and mean pooling already captures most of their signal as a 1-bin low-pass.

The testable prediction is that as the task timescale shortens, the best-probe band should move upward. Per-frame chord detection (Table 1, last block) is one such shorter-timescale case, but absolute accuracies are low across the board, so it is not yet decisive. A cleaner test would pair one audio dataset with tasks at controlled timescales—genre vs. onset detection on the same clips, or synthetic stitch-and-shuffle tasks in which the label timescale is varied while content statistics are held fixed—and read off how the best band moves with task timescale.

5.4 Looking Ahead

If spectral concentration holds across standard MIR labels, one practical implication is that frozen-encoder inference may only need the lowest few rFFT bands (a fixed temporal low-pass plus down-sampling) rather than the full (F, T) representation, with little or no accuracy loss. We do not test this claim here: the experiments above were designed to probe the phenomenon, not to validate a compression scheme. We flag it as a natural baseline for future work, to be tested alongside the three hypotheses above.

6. ACKNOWLEDGMENTS

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A. EXPERIMENTAL SETUP – DETAILS

This appendix records implementation details that are not needed to read the main results but are required to reproduce them. All encoders are loaded from public checkpoints and used with no fine-tuning; linear probes are ℓ_2 -regularised softmax multinomial logistic regressions. Unless stated otherwise, each task is evaluated with a class-stratified 80:20 train/test split on its working pool, and accuracies in Table 1 are classification accuracy on the held-out test split.

A.1 Model Checkpoints and Feature Shapes

For every encoder we report the checkpoint, how each input clip is prepared, and the exact (F, T) shape of the per-channel, per-frame output that feeds Eqn (1). F is the number of feature channels, T the number of time frames, and $K = \lfloor T/2 \rfloor + 1$ the number of one-sided DFT bins per channel.

The audio branch of CLAP [9] is HTS-AT [16], which processes 10-second clips at 48 kHz mono and returns a last-hidden-state tensor of shape $(1, 768, 2, 32)$: 768 latent channels and a 2×32 post-swin token grid (two frequency-patch slots, 32 time frames). To cover music-length clips we encode two non-overlapping 10-second segments per track—seconds $[0, 10)$ and $[10, 20)$, zero-padding the second segment if the track is shorter than 20s—and concatenate the two last-hidden-states along time, yielding $(1, 768, 2, 64)$. Before probing, the leading two-patch frequency axis is merged into the channel dimension by stacking the two patch slots contiguously, $(B, 768, 2, 64) \rightarrow (B, 1536, 64)$, so that $F=1536$, $T=64$, $K=33$. This merge is the reason f in Eqn (1) indexes channel-by-frequency-patch pairs rather than raw latent channels for CLAP.

For AudioMAE we use the publicly released ViT-B/16 checkpoint pretrained on AudioSet-2M and fine-tuned on AS-20k [10]. Inputs are Kaldi log-Mel spectrograms at 16 kHz mono with 128 Mel bins, 10 ms frame shift, and a Hanning window, normalised with $(x - \mu)/(2\sigma)$ using the published fine-tuning statistics ($\mu = -4.2677$, $\sigma = 4.5690$) and padded or cropped to 1024 frames (10.24 s). ViT-B/16 with patch size 16 produces a 64×8 (time $\times$ Mel) token grid of 768-dimensional tokens; we drop the CLS token and reshape the remaining 512 patch tokens by collapsing the eight frequency patches into the channel dimension, $(B, 512, 768) \rightarrow (B, 64, 8 \cdot 768) \rightarrow (B, 6144, 64)$, giving $F=6144$, $T=64$, $K=33$.

For BEATs we use the official iter-3 checkpoint [11] via the reference implementation. Audio is resampled to 16 kHz mono, and the returned per-frame feature sequence is transposed to an (F, T) layout. On FMA-Small 30-second tracks this yields $T \approx 527$ per clip at the native BEATs frame rate, giving $K=264$. BEATs is therefore the only encoder in our set whose native time resolution exceeds 100 frames per clip, which lets the main table test whether the spectral concentration effect is an artifact of the coarser $T=64$ grid used by CLAP and AudioMAE.

For PaSST we use the PaSST-base AudioSet checkpoint [12] together with its standard timestamp-embedding interface at 32 kHz mono. PaSST returns a timestamp tensor of shape $(B, T_{\text{passt}}, 1295)$, which we transpose to $(B, 1295, T_{\text{passt}})$ so that time is the last axis. T_{passt} is determined by the audio window length: 20 s of FMA audio gives $T=601$ ($K=301$), and 4-second NSynth notes give $T=81$ ($K=41$). With band width $W=4$ this yields $B=75$ bands on FMA and $B=10$ bands on NSynth.

A.2 CLAP Architectural Periodicity

CLAP (HTS-AT) and AudioMAE (ViT-B/16) both use a fixed-stride patch grid, which introduces a repeating pattern in the temporal FFT. For CLAP, after concatenating two 10-second windows the resulting $(1536, 64)$ embedding shows spectral peaks at $k=8, 16, 24$ (i.e. every 16 frames out of $T=64$). AudioMAE with patch size 16 and $T=64$ exhibits the same peaks for the same reason. PaSST uses randomised patchout during training and does not show this pattern.

Figures 3–5 document this pattern from three angles: a single-sample 3D surface, a population heatmap, and a cross-encoder mean-spectrum comparison. Because this artifact confounds band-level interpretation for CLAP and AudioMAE, the four null-perturbation experiments were run on PaSST only.

Single-sample 3D dimxfreq surface (genre=Electronic, sample_idx=1233)

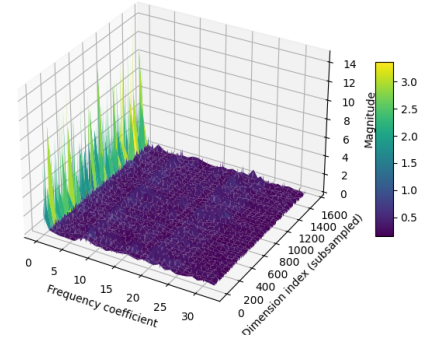


Figure 3. $|\text{rFFT}|$ surface of one CLAP embedding (Electronic, idx 1233). Spikes at $k=8, 16, 24$ appear across all 1536 dimensions.

A.3 Task Label Inventories and Splits

All four tasks are closed-set multi-class classification and use the same softmax probe of Eqn (5).

FMA-Small genre is an eight-way classification over the top-level FMA genres {Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock} [13]. One embedding is computed per track over the encoder’s full input window (20 s for CLAP and PaSST, 30 s for BEATs, 10.24 s for AudioMAE). The CLAP probe uses all 7 997 tracks of the balanced set (three files are excluded for decoding errors); PaSST, AudioMAE, and BEATs use the same genre-stratified 800-track subsample (100 tracks per

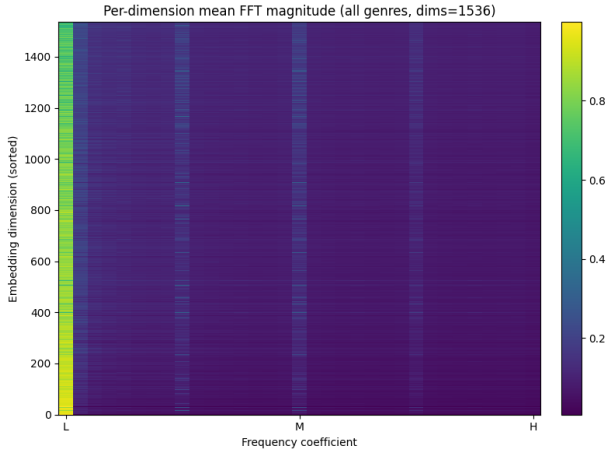


Figure 4. Per-dimension mean $|\text{rFFT}|$ for CLAP over all 7 997 FMA-Small tracks. Bright columns at $k=8, 16, 24$ are sample-independent.

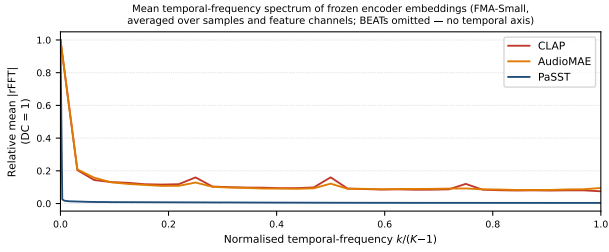


Figure 5. Mean temporal-frequency spectrum (DC-normalised) for CLAP, AudioMAE, and PaSST on FMA-Small. CLAP and AudioMAE share the same patch-grid harmonics; PaSST is DC-dominated with no periodic artifact.

genre). Splits are class-stratified 80:20 with a fixed random seed.

NSynth instrument-family classification is an eleven-way problem over the canonical family set {bass, brass, flute, guitar, keyboard, mallet, organ, reed, string, synth-lead, vocal} [15]. We concatenate the official validation and test partitions (up to 16 774 notes), encode one embedding per 4-second note, and re-split 80:20 stratified by instrument family; each encoder uses the subset of notes it successfully encoded, which ranges from 12 678 (PaSST) to the full 16 774 (CLAP, BEATs).

GuitarSet key detection uses all 360 mono pickup-mix tracks across the six players, five playing styles (BN, Bossa Nova; Funk; Jazz; Rock; SS, Singer-Songwriter), and both comp (accompaniment) and solo takes [14], with the track-level JAMS `key_mode` annotation as the label. Labels are of the form root-major or root-minor over the 12 chromatic roots, a 24-way space in principle; the 360-track pool actually populates 19 of those 24 classes. One embedding per track, keeping the encoder’s full (F, T) output, becomes one sample, and the probe is evaluated with an 80:20 class-stratified split.

GuitarSet chord detection uses the same 360 mono pickup-mix tracks and the same 80:20 split pattern (but

stratified on the per-track majority chord rather than key) for a per-frame 25-way classification. Labels are taken from the first (simple maj/min) JAMS chord annotation, reduced with the MIREX protocol to $\{C, C^\sharp, \dots, B\} \times \{\text{maj}, \text{min}\}$ plus a no-chord class, and aligned to each encoder’s frame grid by locating each frame-centre time $(i+0.5) \cdot \Delta t$ in the JAMS interval timeline ($\Delta t=0.312$ s for CLAP, 0.160 s for AudioMAE, 0.100 s for PaSST, and ~ 0.020 s for BEATs at its native per-frame rate). Because the pre-extracted BEATs embeddings used for key detection are the model’s mean-pooled 527-class AudioSet logits—which have no usable time axis—we re-extract per-frame BEATs features for the chord block by bypassing BEATs’ classification head and returning the pre-pool transformer outputs, yielding $(F=768, T=1496)$ per track at the native BEATs patch rate of ~ 50 Hz.

A.4 Datasets and Pool Sizes

FMA-Small is the eight-class balanced subset of the Free Music Archive: 1 000 tracks per genre, 30 s per track, 8 000 tracks total [13]. Our working pool is 7 997 tracks after dropping three files with decoding errors.

NSynth is the isolated-note dataset of Engel et al. [15], with each note 4 s at 16 kHz. We use the official validation and test partitions concatenated, giving up to 16 774 notes across the 11 instrument families; per-encoder subsets are listed in Table 1.

GuitarSet is a 360-excerpt solo-guitar dataset by six players across five styles with two microphone mixes, each with synchronised JAMS annotations [17] for chord, beat, key, and pitch [14]. We use the full 360-track mono pickup-mix pool (all six players, all five styles, and both comp and solo takes) and take the track-level JAMS `key_mode` annotation as the supervised label.